

The Verification Gap: Why AI-Augmented Academic Writing Needs Reporting Guidelines for Reasoning

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Abstract

Large language models are increasingly used to assist with academic writing. Citation hallucination has received considerable attention, but AI writing assistants also create verification challenges that existing tools do not address: confidence inflation, scope creep, and fluent prose that hides gaps in reasoning. A gap in reporting guidelines compounds the problem. The EQUATOR Network maintains nearly 700 guidelines for empirical health research, yet to our knowledge no consensus-based equivalent exists for non-empirical paper types. Drawing on Gregor’s (2006) theory-type taxonomy, we identify two non-empirical theory types for which argument quality, more than factual accuracy, is the primary verification challenge. We propose a typed verification model that distinguishes claims (source-verifiable facts), arguments (reasoning chains checked with Toulmin’s model), and propositions (recommendations with boundary conditions, checked with Whetten’s criteria), each with its own procedure. We propose that the scholarly community develop verification infrastructure for non-empirical papers, analogous to what EQUATOR provides for empirical research.

Keywords: academic writing, artificial intelligence, reporting guidelines, verification, argumentation, non-empirical research

Key Points:

- AI writing tools create verification challenges beyond citation hallucination, including confidence inflation and argument degradation.
- EQUATOR provides nearly 700 guidelines for empirical research; we found no consensus-based equivalent for non-empirical papers.
- We propose typed verification that distinguishes claims, arguments, and propositions, each with its own procedure.
- Applying the wrong verification procedure to a unit type may produce false failures.

1 Introduction: The Problem

Writing assistants based on large language models (LLMs) have spread quickly: up to 22% of computer science papers now show signs of LLM editing, and such editing can be detected in every major discipline (Liang et al., 2025). The tools write fluently, and they can synthesise literature, structure an argument, draft prose, and format references. That fluency brings risks of its own, and some are harder to spot than the problems of writing without AI. When a human author falls short, the text tends to show it, in awkward phrasing, gaps in coverage, or inconsistent formatting. An AI-assisted draft can read as polished and complete while it contains errors that only checking would reveal, such as a citation to a paper that does not exist.

The most widely discussed risk is citation hallucination. LLMs can invent plausible-sounding papers, complete with fabricated authors, journal names, and digital object identifiers (DOIs). Mugaanyi et al. (2024) found that 62–89% of DOIs generated by large language models for natural-science and humanities topics were fabricated, with DOI accuracy as low as 8.5% in some disciplines. Walters and Wilder (2023) found fabricated citations in 55% of GPT-3.5 references and 18% of GPT-4 references in generated literature reviews, lower for the newer model but still substantial. A hallucinated citation is a distinctive kind of falsehood: unlike a factual error that contradicts known evidence, it cannot be caught through domain knowledge alone, because the work it cites does not exist. Citation hallucination has received considerable attention, and tools designed to detect it are emerging.

Citation hallucination may, however, be only the most visible failure mode. A second risk, which may be harder to notice, is what we term *confidence inflation*: AI writing assistants may express more certainty than the evidence warrants. LLMs verbalise high confidence even on questions they answer wrongly (Xiong et al., 2024; Zhou et al., 2024), and LLM summaries of scientific work generalise findings beyond what the original studies claim (Peters and Chin-Yee, 2025). Whether this reaches academic prose is less clear. In one corpus of applied-linguistics abstracts, ChatGPT used boosters such as “shows” at twice the human rate, but also more hedges (El-Dakhs et al., 2025), and no study we know

of yet measures boosters against the strength of the evidence. The stakes are epistemic. “Demonstrates” and “suggests” tell readers whether a claim rests on strong evidence or on preliminary findings, so inflated prose can misrepresent the state of the evidence without containing a single factually incorrect statement.

A third risk applies specifically to technical and empirical papers: AI-generated equations and derived values can contain arithmetic errors that survive review because the results look plausible. A number in a physically reasonable range can pass a plausibility review even when the arithmetic behind it is wrong. This suggests that for quantitative content, *mechanical reproduction* (substituting values and recomputing) catches errors that *expert assessment* (judging whether the physics looks right) does not.

A fourth risk is scope creep. Without structural constraints such as section specifications and word budgets, AI-assisted drafts may stretch arguments beyond what the available evidence supports, add tangential sections, and exceed format requirements. Scope is hard to manage in all academic writing, but AI tools can make it harder: they generate content with little effort, so expansion becomes the path of least resistance.

None of these failure modes can be detected by checking whether citations are real. Tools that check automatically whether references exist, and model-level approaches such as retrieval-augmented generation (in which the model is given access to source documents while it writes), address the factual accuracy of AI-assisted writing. Some recent systems go further. A linter checks a manuscript’s references, internal consistency, and causal language against the study design (Samsonau, 2026); a workflow pairs each drafted claim with an evidence packet and sends unsupported claims back to the author (Chen et al., 2026); a provenance discipline binds every reported quantity to a sealed trail (Zhou and Yu, 2026); and a checklist that verifies cited assertions in narrative reviews is in development (VANRA Development Group, 2026). Each asks whether evidence exists, supports a claim, or suffices. To our knowledge, none distinguishes claims from arguments and propositions and applies a procedure suited to each. That gap matters most for papers whose main contribution is an argument.

AI-text detection is no substitute: it asks who wrote a text, and verification asks

whether the text holds. Detectors are also fragile: their accuracy varies with generation settings (Dugan et al., 2024), and paraphrase can collapse it (Krishna et al., 2023).

The failure modes described above arise in the writing process more than in the AI model. They occur when, as far as we know, nothing checks whether the language of an argument or proposition matches its support, or whether an argument is logically sound and not only well-sourced. This paper asks what such infrastructure might look like for arguments and propositions, and whether existing scholarly frameworks provide a foundation for building it.

2 The Landscape Gap

The most comprehensive verification infrastructure for academic writing to date is the EQUATOR Network. EQUATOR (Enhancing the QUALity and Transparency Of health Research) maintains a database of nearly 700 reporting guidelines for health research (EQUATOR Network, n.d.). These guidelines, among them CONSORT for randomised controlled trials, STROBE for observational studies, and PRISMA for systematic reviews (Page et al., 2021), give authors structured checklists to complete while preparing a manuscript. The GoodReports tool (Struthers et al., 2021) selects the guideline that fits a study type, which lowers the barrier to adoption further. A Cochrane review found that journals endorsing CONSORT favoured completeness on 25 of 27 checklist items, significantly so for five (Turner et al., 2012), and incomplete reporting has been identified as a major contributor to research waste (Glasziou et al., 2014).

Yet the EQUATOR database points to a structural gap. Virtually all of its nearly 700 guidelines address research that collects, analyses and reports data. Systematic reviews (PRISMA; Page et al., 2021) are a partial exception, since they synthesise primary data without generating it. Beyond them, to our knowledge, no consensus-based guideline of the EQUATOR type exists for theoretical, perspective, or conceptual papers. Conceptual papers lack “commonly accepted templates” (Jaakkola, 2020, p. 18). Design-science researchers have reported no agreed guidance on presenting and reviewing their papers (Gregor and Hevner, 2013; Dinter and Krawatzeck, 2015), and validity in the field “is

neither well-understood nor consistently established” (Larsen et al., 2025). Software engineering’s community standards come closest: the one for methodological papers asks for “clear, valid arguments” but gives no procedure for checking them (Ralph et al., 2020). The best-known narrative-review instrument is “not a reporting guideline” (Baethge et al., 2019), and guidelines are only now being developed for concept analysis (Nashwan et al., 2026) and narrative reviews (VANRA Development Group, 2026). None sets out how to verify arguments or propositions as distinct from cited statements.

This absence may reflect a difference in what needs verifying across paper types. Gregor (2006) identified five theory types in information systems research, based on four goals: analysis, explanation, prediction, and prescription. Two of these types are non-empirical in their primary contribution: Type I (analytic theories, including taxonomies and frameworks) and Type V (design and action theories, including design principles and methods), although Type V papers may include an empirical evaluation of the designed artefact. Their core contributions are conceptual structures, which are valued for clarity, coherence, and utility and cannot be checked against measurements the way data points can. The remaining types (Type II: explanatory, Type III: predictive, Type IV: explanatory and predictive) fit empirical verification more naturally, because they make claims about observable phenomena.

Gregor’s taxonomy was developed for information systems, but the underlying distinction applies more widely. The IS community has developed structured methodologies for non-empirical work, notably Peffers et al.’s (2007) widely adopted design science research process model, yet these methodologies have not been turned into reporting guidelines analogous to EQUATOR. Perspective articles, commentaries, theoretical syntheses, and methodological proposals share the same feature: their primary contribution is an argument or a framework, not empirical evidence. Such papers appear in virtually every academic discipline, from the humanities and social sciences to engineering and medicine.

For these non-empirical paper types, the central verification question may be less “do the data support the claims?” and more “is the argument logically sound, are the premises established, and are the boundary conditions specified?” (Toulmin, 2003; Whetten, 1989).

That is a different task from the one EQUATOR guidelines are designed to support.

This suggests that the gap in EQUATOR’s coverage is structural. Empirical papers need verification of research design, data collection, analysis, and reporting, and structured checklists suit those well. Non-empirical papers need verification of argument quality: whether warrants are valid, reasoning is coherent, confidence is calibrated, and boundaries are specified. These are different verification tasks that may require different infrastructure. The gap is most acute in purely non-empirical papers, but it is a systemic risk wherever authors move from data-driven claims to interpretive reasoning, including the discussion sections of empirical research. The question is whether established frameworks from argumentation theory and theory-building methodology can provide a foundation for this infrastructure.

3 Typed Verification: A Proposal

We propose that verification infrastructure for non-empirical papers should distinguish three unit types, each with its own verification procedure (Table 1).

Table 1: Three verification unit types with per-type procedures.

Type	Definition	Verification question	Framework
Claim	Source-verifiable fact	Does the source say this?	Citation checking
Argument	Evidence + inference $\rightarrow$ conclusion	Is the reasoning valid?	Toulmin (1958/2003)
Proposition	Recommendation with boundaries	Are conditions specified?	Whetten (1989)

The first and most familiar type is the **claim**: a factual statement that can be checked against one or more sources. Claims are the default unit in citation-based verification. “Study X found Y” is a claim, and its accuracy can be checked by reading Study X. Most existing verification tools, including citation checkers and fact-verification systems, work on claims. For a claim the verification question is simple: does the cited source exist, and does it say what the manuscript attributes to it?

The second type is the **argument**: a chain of reasoning that combines evidence with inference to reach a conclusion. Arguments appear throughout academic writing, particularly in discussion and conclusion sections, yet they resist claim-level verification, because

their validity depends on the logical relation between premises and conclusion and not on any single source. Toulmin (2003) provides a well-established framework for analysing argument structure. It decomposes an argument into six components: claim (the conclusion asserted), grounds (the evidence offered), warrant (the inference that connects grounds to claim), backing (support for the warrant), qualifier (the degree of certainty), and rebuttal (conditions under which the argument does not hold). This framework may provide a basis for verifying arguments in academic writing: an argument can be evaluated by checking whether its grounds are established, its warrant is explicit and valid for the target audience, and its qualifier matches the strength of the evidence. (Our operationalisation uses five of the six components and treats backing as implicit when the target audience already accepts the warrant.) Gupta et al. (2024) found that LLMs can extract Toulmin-style argument structures through zero-shot prompting, which suggests that automated support for argument-level verification may be technically feasible.

The third type is the **proposition**: a novel recommendation or contribution that specifies what should be done and under what conditions. Propositions are the defining unit of many non-empirical papers, because they are what such a paper contributes to its field. Whetten (1989) provides a framework for evaluating theoretical contributions through four components: What (the constructs or factors involved), How (the relationships between them), Why (the underlying logic), and Who/Where/When (the boundary conditions that say where the contribution applies). A proposition can be evaluated without empirical data, by checking whether key constructs are defined, the reasoning is explicit, and the boundary conditions are stated.

The three types may benefit from a shared confidence calibration system. We propose mapping confidence tiers to prescribed language (Table 2), so that the certainty expressed in the text follows the strength of the underlying evidence. This mapping may let authors and reviewers detect confidence inflation, the failure mode described in Section 1, during drafting, before peer review.

The central argument of this paper is that these three types require fundamentally different verification procedures, and that applying the wrong procedure to a type may

Table 2: Proposed confidence-to-language mapping.

Tier	Assign when...	Prescribed language
Established	Multiple independent sources; no dispute	“demonstrates,” “confirms”
Supported	2–3 sources agree; open questions remain	“indicates,” “evidence suggests”
Emerging	1–2 sources; not yet replicated	“may,” “preliminary evidence”
Speculative	Logical inference; no direct data	“warrants investigation”

produce systematic errors. A claim evaluated as an argument, by checking its warrant instead of its source, wastes effort. More consequentially, an argument evaluated as a claim, by checking whether it has a source instead of whether its reasoning is valid, may produce false failures: entries scored as unverified because the procedure did not suit their unit type, not because they are wrong. This argument rests on the premise that different epistemic contributions (facts, inferences, recommendations) have different validity criteria; if the premise holds, a single verification procedure will systematically misclassify some entries. One might ask whether a simpler split between empirical and non-empirical verification would do. The distinction between arguments and propositions matters because they fail differently: an argument with a weak warrant needs better reasoning, and a proposition with missing boundary conditions needs better scoping. Putting both under one “non-empirical” category hides which correction is needed.

4 Related Work and Design Rationale

The framework described in Section 3 is proposed as a typed addition to the approaches reviewed in Section 1. An empirical evaluation across several paper projects, authors, and domains is needed before general claims about its effectiveness can be made. Appendix A gives one self-application data point: the verification registry kept while this paper was written, which shows the framework applied prospectively to a perspective paper.

The design draws on independent research on structured verification. Results suggest that representing support and attack relations formally improves LLM review aggregation over end-to-end approaches (Sukpanichnant et al., 2024), and that prompting with Toulmin’s model improves LLM argument extraction over generic prompts (Gupta et al.,

2024). Both suggest that structured frameworks combined with LLMs may outperform LLMs alone, which is consistent with the approach proposed here.

The proposal is offered as a model for the scholarly community to test and adapt. Whether typed verification adds real value across authors, domains, and writing workflows remains an open empirical question.

5 Call to Action

AI writing assistance creates verification challenges beyond citation hallucination (Section 1), yet EQUATOR’s nearly 700 reporting guidelines address empirical research almost exclusively, which leaves non-empirical paper types without systematic verification support (Section 2). On the basis of this gap, and of the argument that claims, arguments, and propositions each need their own verification procedure (Section 3), we propose that the scholarly community develop verification infrastructure for non-empirical papers, with procedures differentiated by unit type:

- **Source verification** for claims (does the cited source say this?);
- **Toulmin-based analysis** for arguments (is the reasoning valid, are the premises established, is the qualifier calibrated?);
- **Whetten-based analysis** for propositions (are the boundary conditions specified, is the underlying logic explicit?).

Each type would have its own checklist, confidence calibration criteria, and language guidelines that authors can apply while preparing a manuscript.

This proposal has specific boundary conditions. It applies to academic papers intended for peer-reviewed publication in fields where argument quality is a primary contribution. It does not apply to informal writing, journalism, creative writing, or very short opinion pieces, where the overhead of formal verification would exceed the benefit. The verification frameworks proposed here, Toulmin’s argument model and Whetten’s framework for theoretical contributions, are grounded in Western academic argumentation traditions.

Their fit with scholarly traditions that have different rhetorical conventions would need to be assessed, not assumed.

One might object that journal peer review already verifies the arguments of non-empirical papers, since reviewers routinely evaluate logical coherence, boundary conditions, and engagement with counter-arguments. The objection has merit, and peer review remains essential. AI writing assistance may, however, stretch the verification challenge beyond what peer review was designed to handle. When AI tools can produce fluent, well-structured prose with little effort, the volume and apparent quality of submitted manuscripts may rise while the share of well-reasoned arguments among them falls. Reviewers already face growing workloads; detecting AI-generated confidence inflation and implicit warrant failures as well may exceed what the system can absorb. Process-level verification infrastructure would complement peer review, not replace it: it would catch argument-level problems before submission, so that reviewers can concentrate on substantive evaluation.

Several directions warrant investigation. First, the typed verification model proposed here has been developed and applied within medical simulation and instrumentation. Discipline-specific verification templates, which adapt the three-type model to the conventions of other fields, would be needed to assess how far the framework generalises. Second, integration with AI writing tools could enable verification during drafting, with the checklists and calibration criteria serving as prompts for writing assistants. Third, an empirical comparison of the writing quality of papers written with and without typed verification would provide the evidence needed to move the proposal from preliminary to supported. Fourth, although this paper focuses on non-empirical paper types, the discussion sections of empirical papers make the same move from data to interpretation, and may benefit from the same typed procedures applied to that step. Fifth, the calculation errors described in Section 1, where AI-generated equations produce plausible but incorrect results, suggest that numerical reproduction (mechanically recomputing derived values from stated inputs) may warrant recognition as a distinct verification procedure, next to source checking, argument analysis, and proposition evaluation.

The EQUATOR Network has spent two decades building nearly 700 reporting guidelines for empirical health research, and these guidelines have measurably improved the quality and transparency of published research (Turner et al., 2012; Glasziou et al., 2014). We know of no consensus-based equivalent for non-empirical papers: theoretical contributions, design science research, perspective articles, and conceptual papers. As AI writing tools make fluent academic prose easier to produce, the gap between what is published and what is rigorously verified may widen. We suggest it is time to start building verification infrastructure for these papers too.

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[To be added.]

A Self-Application of the Typed Verification Framework

This paper uses the framework it describes. Its claim registry was kept during writing: entries were registered, typed, and verified as the paper was drafted. This appendix summarises the registry as a prospective data point that reviewers can verify independently.

Registry summary

Table 3 lists all 17 registered entries. Their distribution, 15 claims, 1 argument, and 1 proposition, reflects the paper’s structure: a perspective article that rests mainly on verifiable facts and reasoning chains and ends in a single recommendation.

Worked examples by type

Claim (S2-1). “The EQUATOR Network maintains nearly 700 reporting guidelines.” This entry was first drafted as “approximately 500 guidelines”, based on a secondary source. Verification against the EQUATOR website (accessed 3 March 2026) returned

Table 3: Complete verification registry for this manuscript (17 entries).

ID	Type	Priority	Confidence	Status
S1-1	Claim	P0	Supported	Verified
S1-2	Claim	P0	Supported	Verified
S1-3	Claim	P1	Emerging	Verified
S1-4	Claim	P1	Emerging	Verified
S1-5	Claim	P1	Emerging	Verified
S1-6	Claim	P2	Supported	Verified
S2-1	Claim	P0	Established	Verified
S2-2	Claim	P0	Emerging	Verified
S2-3	Claim	P1	Emerging	Verified
S2-4	Claim	P1	Emerging	Verified
S3-1	Claim	P1	Emerging	Verified
S3-2	Claim	P1	Emerging	Verified
S3-3	Claim	P1	Emerging	Verified
S3-4	Argument	P0	Emerging	Verified
S4-3	Claim	P1	Emerging	Verified
S5-1	Proposition	P0	Emerging	Verified
S5-2	Claim	P2	Speculative	Verified

699 guidelines, and the text was corrected to “nearly 700.” The original figure would have understated the landscape by approximately 30%, illustrating how even a straightforward factual claim benefits from structured source checking.

Argument (S3-4). “Claims, arguments, and propositions require fundamentally different verification procedures.” This is the paper’s central argument. Table 4 shows the Toulmin decomposition as applied during verification.

Table 4: Toulmin decomposition of the central argument (S3-4).

Component	Content
Claim	Different unit types require different verification procedures.
Grounds	A registry entry classified as a source-backed claim can be scored at low confidence for lacking a citation, even when its underlying reasoning would satisfy a Toulmin-style argument checklist (premises verified, warrant logically valid).
Warrant	If applying the wrong verification procedure to a correctly reasoned entry produces a false failure, then type-specific procedures are necessary to avoid systematic misclassification.
Qualifier	Emerging: argued on design rationale; not independently evaluated across multiple paper projects.
Rebuttal	The argument rests on a structural illustration; a larger sample across domains would be needed to establish generalisability.

Proposition (S5-1). “The scholarly community should develop verification infrastructure for non-empirical papers.” This is the paper’s single proposition. Its Whetten decomposition: *What:* typed verification infrastructure that distinguishes claims, arguments, and propositions. *How:* per-type checklists adapted from Toulmin (arguments) and Whetten (propositions), with a confidence-to-language mapping. *Why:* EQUATOR provides nearly 700 guidelines for empirical research (S2-1), while we found no consensus-based equivalent for non-empirical papers (S2-2); each unit type needs its own verification procedure (S3-4); and AI writing amplifies the verification challenge. *Boundary conditions:* applies to academic papers intended for peer-reviewed publication; excludes informal writing, journalism, and creative writing; the proposed checklists are grounded in Western argumentation traditions and may need adaptation for other scholarly conventions.

Corrections discovered during verification

The verification process caught two kinds of issue before submission: (1) the EQUATOR guideline count was corrected from approximately 500 to nearly 700 (S2-1); and (2) the citation hallucination claim (S1-1) at first lacked a source, which prompted a literature search that found Mugaanyi et al. (2024). These corrections, from a factual error to missing evidence, are the kinds of issue the framework is designed to catch.

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